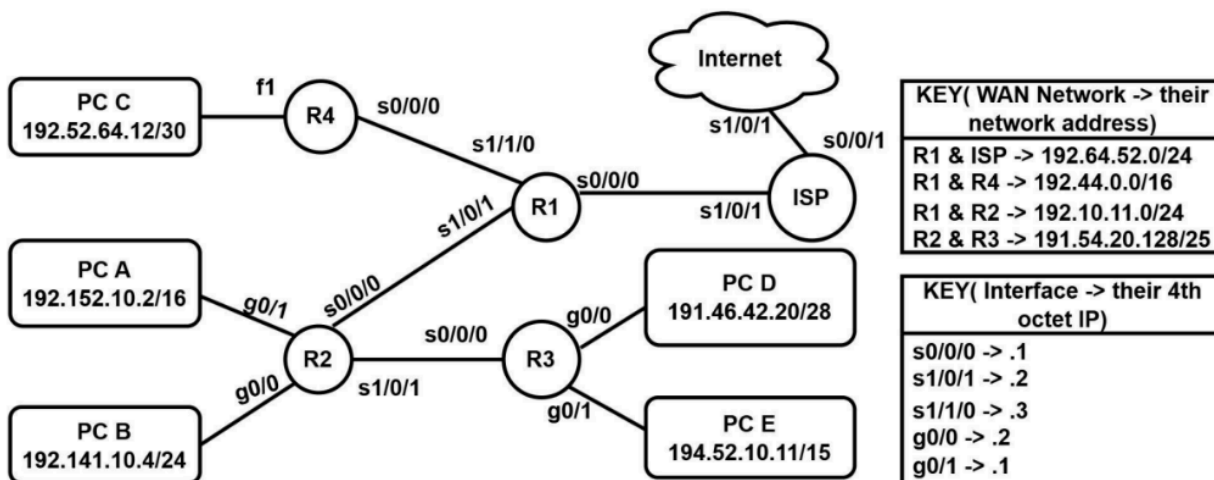


Lecture 11-13 Routing

Spring 25 (A)



- I. On R3, **identify** the networks that will be added to the table without any routing configuration.
- II. **Configure** a directly attached static route on R2 to reach R4 LAN with AD 50
- III. **Configure** a recursive route so that a S* entry is added in R1's routing table. **Identify** the significance of S*.
- IV. **Determine** the AD of a back up route for III .

I. Networks Added to R3 Without Routing Configuration

Directly connected networks are added to the routing table automatically once the interfaces are configured and active.

R3 has three active interfaces:

1. **g0/0 (LAN):** Connected to PC D. The network address is **191.46.42.16/28**. (Note: 191.46.42.20 falls in the .16 subnet).
2. **g0/1 (LAN):** Connected to PC E. The network address is **194.52.0.0/15**.
3. **s0/0/0 (WAN):** Connected to R2. According to the WAN key, the R2 & R3 link is **191.54.20.128/25**.

Alternative Answer: 191.46.42.20, 194.52.10.11, 191.54.20.128

II. Directly Attached Static Route on R2 to reach R4 LAN

A **directly attached** static route uses the outgoing interface rather than the next-hop IP.

- **Target LAN (PC C):** 192.52.64.12/30. The network address is **192.52.64.12** (since /30 has 4 IPs, and .12 is a multiple of 4, this is the network ID). Mask is **255.255.255.252**.
- **Exit Interface from R2 toward R4:** s0/0/0 (via R1).
- **Administrative Distance (AD):** 50.

Configuration Command: R2(config)# ip route 192.52.64.12 255.255.255.252 s0/0/0 50

III. Recursive Default Route for R1

A **recursive route** uses the next-hop IP address. An **S*** entry indicates a **Static Default Route**.

- *Significance of S:** The **S** stands for Static, and the ***** indicates this is a "Candidate Default" route. This route acts as the "Gateway of Last Resort," used when a packet's destination does not match any specific entry in the routing table (usually used to send traffic to the Internet).
- **Next-Hop IP for R1 toward ISP:** The link between R1 & ISP is 192.64.52.0/24. According to the interface key, the ISP's interface (s1/0/1 or s0/0/1) would typically be the next hop. Assuming the ISP is **.2** (since R1 is **.1** or vice versa), we point to that.

Configuration Command: R1(config)# ip route 0.0.0.0 0.0.0.0 192.64.52.2

IV. Administrative Distance for a Backup Route

To create a "Floating Static Route" (a backup route), the Administrative Distance must be **higher** than the primary route.

- The primary static route has a default AD of **1**.
 - To make a backup route for the entry in Part III, I need to assign an AD **greater than 1** (e.g., **5** or **10**). If Part III's route specifically had a custom AD, the backup would simply need to be higher than that number.
-

4. Refer to the topology given above, all the routers are running Distance Vector Routing Algorithms. The routers are supposed to share their information with their neighbors. **Determine** how R2 will know about its neighbors to send its information. In the first iteration, **find out** what kind of information R2 will send.

How R2 Finds Its Neighbors

Under the DVR algorithm, R2 identifies its neighbors through a **neighbor discovery process** (typically by listening for "Hello" packets or periodic routing updates on its active interfaces). Based on the provided diagram and key, R2 is directly connected to four other nodes:

Neighbor	Interface on R2	Connecting Network	R2's Local Interface IP
R1	s0/0/0	192.10.11.0/24	192.10.11.1
R3	s1/0/1	191.54.20.128/25	191.54.20.130 (Note: .2 octet)
PC A	g0/1	192.152.10.0/16	192.152.10.1
PC B	g0/0	192.141.10.0/24	192.141.10.2

Information R2 Sends in the First Iteration

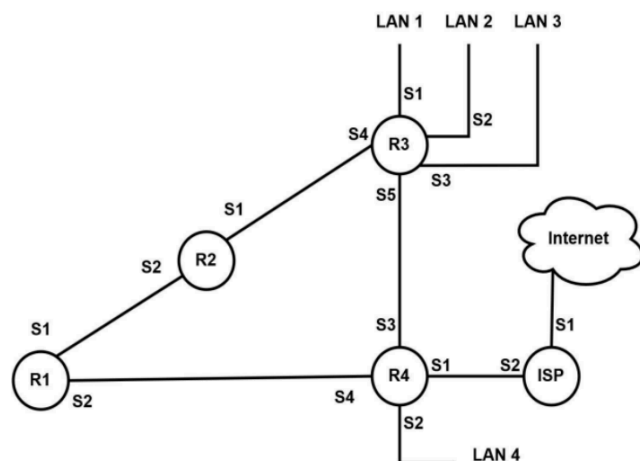
In the **first iteration (Initialization)**, R2 creates an initial distance vector (routing table) containing only its **directly connected networks**. It assumes a cost of **0** for these networks and **infinity** for everything else.

R2 will broadcast/multicast its **Distance Vector** to its neighbors (R1 and R3). The packet it sends will contain the following information:

Destination Network	Cost (Metric)
192.152.10.0/16 (PC A's Network)	0
192.141.10.0/24 (PC B's Network)	0
192.10.11.0/24 (Network to R1)	0
191.54.20.128/25 (Network to R3)	0
<i>All other networks (e.g., PC C, PC D, PC E)</i>	infinity

Key takeaway for the first iteration: R2 does not yet know about the "Internet," PC C, PC D, or PC E. It only tells R1 and R3, "I am the gateway to these four specific subnets." It will only learn about the distant networks in the *second* iteration, after receiving updates from R1 and R3.

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Device	Interface	IP	Network
R1	S1	.101	192.168.10.100/30
	S2	.225	192.168.11.224/30
R2	S1	.193	192.168.12.192/30
	S2	.102	192.168.10.100/30
R3	S1	.1	192.168.96.0/24
	S2	.1	192.168.64.0/25
	S3	.1	192.168.80.0/26
	S4	.194	192.168.12.192/30
	S5	.97	192.168.10.96/30
R4	S1	.1	192.168.9.0/30
	S2	.1	192.168.72.0/27
	S3	.98	192.168.10.96/30
	S4	.226	192.168.11.224/30
ISP	S1	.1	210.1.1.0/24
	S2	.2	192.168.9.0/30

Given the following topology where R1-R4 denote routers and the respective IP table of the topology.

- I. Configure a **recursive** static route in the **R4** router to reach **LAN 1** with AD **2**
- II. Configure the **back-up** route of the above static route configured in **R4**, but this time using exit interface.

I. Recursive Static Route on R4 to LAN 1

A **recursive** static route uses the **next-hop IP address** instead of an exit interface. This requires the router to perform a second lookup in the routing table to find which interface reaches that next-hop IP.

- **Destination Network:** 192.168.96.0
- **Subnet Mask:** 255.255.255.0 (from /24)
- **Next-hop IP (R3's S5 interface):** 192.168.10.97
- **Administrative Distance (AD):** 2

Configuration Command: `ip route 192.168.96.0 255.255.255.0 192.168.10.97 2`

II. Backup Route on R4 using Exit Interface

A **backup route** (also known as a Floating Static Route) must have a **higher Administrative Distance** than the primary route so it only enters the routing table if the primary path fails. Since the primary AD is 2, we will use an AD of 3 or higher. This route will use the **exit interface** as requested.

- **Destination Network:** 192.168.96.0
- **Subnet Mask:** 255.255.255.0
- **Exit Interface (R4's interface toward R1):** S4
- **Administrative Distance (AD):** 3 (Any value > primary AD)

Configuration Command: `ip route 192.168.96.0 255.255.255.0 S4 3`

5. Referring to the above topology, R1, R2 and R3 are running Link State protocol and R1, R4 and the ISP router are running Distance Vector Protocol. **Determine** which routers will send hello packets and **why**. Also, **state** which routers will periodically send routing updates and **why** it is inefficient.

1. Which routers will send "Hello" packets?

Routers R1, R2, and R3 will send Hello packets.

- **Reason:** These routers are running a **Link State Protocol**. In Link State protocols, routers use Hello packets to:
 - Discover neighboring routers on the same link.
 - Establish and maintain "adjacencies" (confirming the neighbor is still alive and reachable).
 - Ensure that communication is bidirectional before exchanging detailed topology information.

2. Which routers will periodically send routing updates?

Routers R1, R4, and the ISP router will periodically send routing updates.

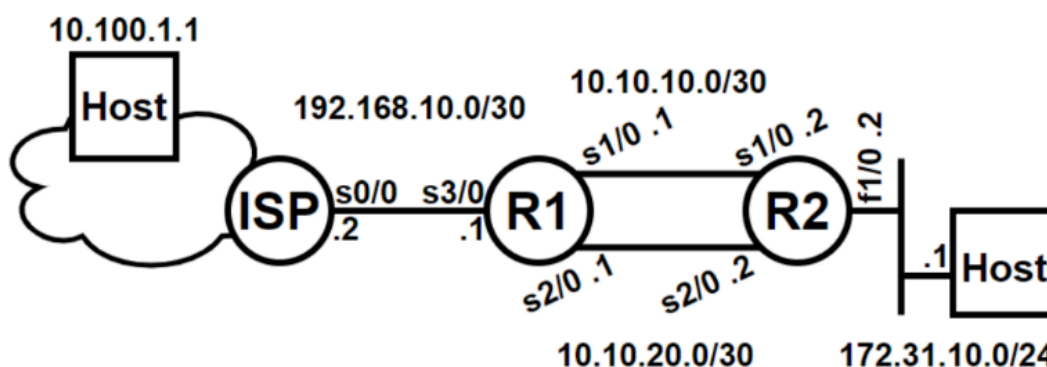
- **Reason:** These routers are running a **Distance Vector Protocol** (such as RIP). Unlike Link State protocols, which only send updates when a change occurs, Distance Vector protocols are designed to broadcast or multicast their entire routing table to their neighbors at fixed intervals (e.g., every 30 seconds for RIP), regardless of whether the network topology has changed.

3. Why periodic updates are inefficient

Sending routing updates on a fixed timer is considered inefficient for several reasons:

- **Bandwidth Consumption:** The routers consume network bandwidth to transmit the same information repeatedly. Even if the network is perfectly stable, the updates continue, which can "clog" slow serial links.
 - **CPU Overhead:** Every time an update is received, the neighboring router must process the packet and re-examine its routing table, wasting processing cycles.
 - **Scalability Issues:** As the network grows, the size of the routing table increases. Sending large tables every few seconds leads to significant overhead, making it difficult for the protocol to scale to very large environments.
 - **Slow Convergence:** Because updates happen on a timer, a failure might occur right after an update was sent. The network then has to wait for the next timer cycle to pass before the "bad news" about the failure is propagated, leading to slower convergence compared to the "triggered updates" used by Link State protocols.
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Refer to the topology above. Since this is a stub network, static routes are preferable. The following static route command was given in the R1 router

```
ip route 172.31.10.0 255.255.255.0 10.10.10.2
```

- State the problem of this command and how to improve it.
 - If the network administrator wishes to configure default static routes, in which routers should s/he configure them? **Justify** your answer.
-

I. Problem with the Static Route Command

The Problem:

The current command `ip route 172.31.10.0 255.255.255.0 10.10.10.2` only utilizes the top link (s1/0) between R1 and R2.

1. **Lack of Redundancy:** If the link 10.10.10.0/30 fails, R1 will lose connectivity to the 172.31.10.0/24 network, even though a physical backup path exists through 10.10.20.0/30.
2. **Underutilization:** The second link (s2/0) remains idle, meaning you aren't benefiting from the available bandwidth of the dual-link setup.

How to Improve it: To make the network more resilient, you should implement a Floating Static Route or Load Balancing.

- **For Load Balancing:** Add a second route with the same administrative distance so the router can use both paths.
 - `ip route 172.31.10.0 255.255.255.0 10.10.10.2`
 - `ip route 172.31.10.0 255.255.255.0 10.10.20.2`
- **For Redundancy (Floating Static):** Add a second route with a higher administrative distance (e.g., 10) so it only activates if the primary fails.
 - `ip route 172.31.10.0 255.255.255.0 10.10.20.2 10`

II. Configuration of Default Static Routes

The network administrator should configure default static routes on **R1 and R2**.

1. On Router R2

- **Configuration:** `ip route 0.0.0.0 0.0.0.0 10.10.10.1` (and/or `10.10.20.1`)
- **Justification:** R2 is at the edge of the stub network. It doesn't need to know every specific route in the outside world or the ISP's network. By pointing a default route toward R1, any traffic destined for the internet or other corporate branches is automatically forwarded upstream.

2. On Router R1

- **Configuration:** `ip route 0.0.0.0 0.0.0.0 192.168.10.2`
- **Justification:** R1 is the gateway to the ISP. Instead of maintaining a massive routing table of internet IP addresses, R1 simply needs to send all non-local traffic to the ISP's next-hop address. This is the standard practice for a stub connection to the internet.

Note: A default route is NOT recommended for the **ISP router** pointing toward R1, as the ISP needs specific routes to reach your internal networks, and pointing a default route back could create a routing loop.

7.State why Distance Vector routing protocol is a **decentralized protocol**, and state at least **two differences** between Distance Vector routing protocol and Link State routing protocol.

Why Distance Vector is a decentralized protocol:

The **Distance Vector (DV)** routing protocol is *decentralized* because **each router makes routing decisions independently** based only on:

- Information from its **direct neighbors**, and
- Its own local routing table (distance vector).

There is **no central controller** or global view of the entire network. Routers periodically exchange their distance vectors with neighbors and update routes using algorithms like **Bellman-Ford**.

Differences between Distance Vector and Link State routing protocols:

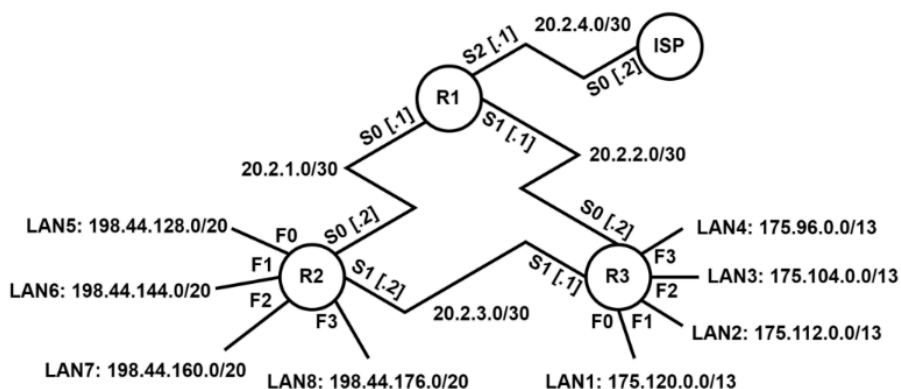
Feature	Distance Vector	Link State
Network View	Topology knowledge from the neighbor point of view	Common and complete knowledge of the network topology
Best Path Calculation	Based on fewest number of hops	Based on the link cost
Updates	Full routing table (sent periodically)	Link State Updates (triggered by changes)

Algorithm	Bellman-Ford	Dijkstra (SPF)
CPU and Memory	Low utilization	Intensive
Hierarchical Structure	No	Yes (e.g., Areas in OSPF)
Convergence Time	Moderate to Slow	Fast

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2. Consider the following figure, where R1-R3 and ISP are routers and the corresponding interfaces are mentioned beside them. The notation S0 [.1] denotes that S0 is the name of the interface and [.1] denotes the IP address. Answer the questions below:

- I. **Calculate** the summarized network of all R2 LANs and then **configure** a summary static route in the router R1 for the calculated network. **(not included in syllabus)**
- II. A default route has been configured in R3 towards ISP via its S1 interface. Now write a **command** to configure a floating default route in R3 towards ISP.



I. Route Summarization for R2 LANs

To summarize the networks, we look for the common bits in the IP addresses of the LANs attached to R2.

1. Identify the Networks

The R2 LANs are:

- **LAN5:** 198.44.128.0 /20
- **LAN6:** 198.44.144.0 /20
- **LAN7:** 198.44.160.0 /20
- **LAN8:** 198.44.176.0 /20

2. Binary Comparison of the Third Octet

Since the first two octets (198.44) are identical, we only need to compare the third octet:

- **128:** 1 0 0 0 0 0 0 0
- **144:** 1 0 0 1 0 0 0 0
- **160:** 1 0 1 0 0 0 0 0
- **176:** 1 0 1 1 0 0 0 0

The first **four bits** (highlighted: 1 0 _ _) are common to all four networks.

- The original prefix was /20.
- By summarizing these four blocks, we move the mask back. The common bits are in the 128, 64, 32, 16 positions. However, looking at the range, they all share the 128 and 64 bit status (1 and 0 respectively).
- The range spans from 128 to 191. The common prefix length for these four /20 networks is 18.

Summarized Network: 198.44.128.0 /18 **Subnet Mask:** 255.255.192.0

3. Configuration on R1

To reach R2 from R1, the next-hop IP address is R2's S0 interface: 20.2.1.2.

Command: Router(config)# ip route 198.44.128.0 255.255.192.0 20.2.1.2

II. Floating Default Route for R3

A **floating default route** is a backup route with a higher Administrative Distance (AD) than the primary route. It only appears in the routing table if the primary link fails.

1. Analyze the Path

- **Primary Path:** Configured via S1 towards ISP (likely through R2).
- **Backup Path:** To reach the ISP from R3 if S1 fails, the traffic must go through interface **S0** towards R1.

2. Configuration Command

The default AD for a static route is 1. To make it "floating," we assign it a value higher than 1 (e.g., 10). The next hop for R3's S0 interface is R1's S1 interface: **20.2.2.1**.

Command: `Router(config)# ip route 0.0.0.0 0.0.0.0 20.2.2.1 10`

(Note: The "10" at the end is the administrative distance that makes it a floating route.)

6. Refer to the figure in above. All routers are running the Link State Routing protocol. How does router R3 know that it needs to send the LSP packets to only its interfaces of S1 and S0? **Explain briefly.**

In Link State Routing protocols, routers use a mechanism called **flooding** to ensure every router has an identical map of the network.

Router R3 knows to send its Link State Packets (LSPs) specifically to interfaces **S1** and **S0** based on two primary rules of the protocol:

1. Neighborhood Discovery (Adjacency)

Before flooding LSPs, R3 sends out "Hello" packets on all its interfaces to discover neighbors.

- **Active Neighbors:** R3 only sends LSPs to interfaces where it has successfully established an **adjacency** with another router.
- **The S1/S0 Links:** In the provided diagram, R3's interfaces **S1** (connected to R2) and **S0** (connected to R1) have active router neighbors. Therefore, it identifies these as the paths to propagate topology information.

2. Interface Type (LAN vs. Point-to-Point)

Link State protocols distinguish between **routing interfaces** (links to other routers) and **stub/passive interfaces** (links to end-user networks).

- **LAN Interfaces (F0, F1, F2, F3):** These interfaces connect to local area networks (LAN1, LAN2, LAN3, LAN4). Since there are no other routers on these segments to process or pass on link-state data, R3 treats them as "leaves" in the network tree.
- **Flooding Logic:** R3 floods LSPs out of all interfaces that are part of the routing process **except** the one on which the information was received. If R3 is originating the LSP, it sends it to all neighbors (S1 and S0). It does not "flood" the LSPs onto the LAN interfaces because there are no routing peers there to receive them.

Summary

R3 sends LSPs to **S1 and S0** because those are its only interfaces connected to **other routers** (R2 and R1) running the same Link State protocol. The LAN interfaces are terminal points for data, not transit points for routing updates.